

TECHNICAL SPECIFICATION

DOCUMENT 00

Ultra-Safe Central Capsule

Scalable Survival Module for Ground, Aerial & Maritime Platforms

Document Version	Date	Status
1.6 Full Detailed Engineering Baseline (6 Independent Safety Layers)	September 2026	Authoritative Full Baseline for Design, Simulation, Testing, Patent Extraction and Licensing

This document establishes the complete and authoritative technical baseline for the Ultra-Safe Central Capsule (Large Configuration). It defines locked geometry, controlling mass properties, structural architecture, the six independent safety layers in full detail, parachute and host-separation systems, impact energy management, life-support and environmental systems, flotation and self-righting capability, emergency location, fire protection, host-platform interfaces, verification strategy, scalability provisions, and preliminary patent-family assessment. All numerical values marked as controlling targets or hard ceilings are frozen for downstream detailed design unless formally revised through configuration control. This document is intended to serve as the single controlling reference for all subsequent detailed design, analysis, simulation, testing, intellectual-property extraction and long-term licensing activities.

Revision Note (Version 1.6 Full Detailed Engineering Baseline): This revision expands all technical descriptions to full detail without summarization, locks eight progressive landing legs, defines a clear centre-of-gravity envelope, specifies the parachute cluster as eight primary plus eight auxiliary canopies, introduces explicit scalability language for a family of capsule sizes, and records that future variants equipped with concealed rotors providing limited hover capability may be designed and engineered if requested by a licensee. The six independent safety layers, continuous-floor carbon-fiber roll-cage philosophy, multi-path separation architecture, door configuration (two side doors plus one large reinforced roof hatch), and seven-person circular/radial seating layout remain locked core features. This version is suitable as the controlling reference for a complete engineering document tree and for patent extraction.

1. Purpose and Design Intent

The Ultra-Safe Central Capsule is an independent, modular survival cell designed for installation at the structural heart of a wide range of host platforms. These host platforms include, but are not limited to, ground vehicles of various classes, helicopters, fixed-wing aircraft, aerial work platforms and cranes, and large maritime structures such as the Fast Ocean Pyramid family and related offshore or amphibious platforms. The single overriding requirement of the design is that the occupants remain alive and recoverable after the complete loss of the host vehicle, whether that loss is caused by high-energy impact, uncontrolled fire, explosion, structural break-up, uncontrolled descent, or submersion in water.

The capsule is deliberately not conceived as a conventional escape pod that relies on a single recovery method. Instead it implements a multi-layer, multi-domain survival architecture that remains functional even when several critical systems have failed simultaneously. After separation from the host the capsule is completely independent of the host for power, control, life support, location and recovery. This functional independence is the foundation of the entire safety philosophy and is a locked design principle.

1.1 Core Design Principles (Locked)

- 1. Functional Independence** – After separation the capsule requires no services from the host vehicle. All power, control, life support and location functions are fully self-contained within the capsule envelope.
- 2. Six Independent Safety Layers** – Every critical survival function is protected by six independent paths. No single-point failure, whether electrical, mechanical, pyrotechnic or structural, can disable the overall survival sequence.
- 3. Structural Continuity** – Geometry and structure preserve a continuous high-strength carbon-fiber roll-cage and an unbroken floor plane that maximises strength and stiffness for the landing system under multi-axis loading.
- 4. Multi-Axis Integrity** – The structure and energy-absorption system are designed for extreme combined loads including vertical impact, lateral impact, roll-over and full inversion.
- 5. End-to-End Survival Sequence** – The design addresses the complete chain from detection of the emergency condition, through clean separation from the host, controlled descent, survivable landing, and long-duration post-landing support and location.
- 6. Certification Orientation** – Architecture, documentation and verification planning are structured from the outset to support both aviation and maritime certification pathways.

2. Geometry, Occupancy and Mass Properties

The large-configuration capsule is defined as a circular-planform cylinder with locked external dimensions of 5.80 m diameter and 3.80 m overall height, the height dimension including the stowed landing-leg envelope. After subtraction of primary structure, secondary structure, systems and installed equipment, the internal usable volume is approximately 55–62 m³. This volume has been assessed as sufficient to accommodate the continuous six-axis carbon-fiber roll-cage, seven radial seats with energy-absorbing suspensions, a clear central access corridor, pilot and co-pilot control stations, emergency equipment stowage and practical emergency egress paths for all seven occupants.

Seven occupants are accommodated in a circular/radial seating arrangement. The pilot and co-pilot positions are oriented for optimal access to the primary control and emergency interfaces while the remaining five seats are distributed symmetrically. The radial configuration was selected after comparative evaluation of linear, facing and radial layouts. It provides the best combination of structural symmetry about the vertical axis, more even distribution of multi-axis acceleration loads among the occupants, and practical access for boarding, in-mission communication and emergency movement within the capsule.

Parameter	Value	Status / Notes
External Diameter	5.80 m	Locked – circular planform
External Height	3.80 m	Locked – includes stowed leg envelope
Internal Usable Volume	55–62 m ³	After structure + systems; packaging validated for 7 occupants + roll-cage
Occupant Capacity	7 persons	Pilot + Co-pilot + 5 passengers
Seating Arrangement	Circular / Radial	Locked – structural symmetry and multi-axis G distribution
Controlling Operational Mass	22 000 kg	Hard ceiling – all systems + 7 occupants + full consumables
Structural Safety Factor	≥ 2.8	Primary roll-cage and continuous floor under ultimate combined loads
Centre-of-Gravity Envelope	Within 120 mm of geometric centre	Vertical and horizontal; mandatory for stable descent and landing
Number of Openings	3	2 side doors + 1 reinforced roof hatch; continuous floor (no floor hatch)
Landing Legs	8	Evenly spaced around circumference; progressive hydraulic + elastomer absorption

2.1 Controlling Mass Budget

The mass budget presented below is the controlling allocation for all detailed design work. Any projected growth beyond the hard ceiling of 22 000 kg requires formal baseline change control and may necessitate resizing of the parachute cluster and the landing-leg energy-absorption system. A deliberate contingency margin is retained within the budget to absorb normal design evolution without immediate violation of the hard mass ceiling. Component-level masses will be refined through detailed CAD and supplier data, but the total shall not exceed 22 000 kg without formal revision of this document.

Group	Allocation (kg)	Share
Primary structure & continuous roll-cage	6 300 – 6 600	≈ 29.5 %
Landing legs (8) + dampers + pads	2 600 – 2 800	≈ 12.5 %
Parachute system (8 primary + 8 auxiliary)	1 450 – 1 600	≈ 7 %
Separation actuators & interface hardware	950 – 1 050	≈ 4.5 %
Energy storage & electrical systems	1 850 – 2 050	≈ 9 %
Life support + 72 h consumables	1 150 – 1 250	≈ 5.5 %
Flotation, self-righting, location systems	900 – 1 000	≈ 4.5 %
Fire suppression, sealing, interiors	1 100 – 1 200	≈ 5.5 %
Seats, harnesses, human-factors equipment	980 – 1 080	≈ 5 %
Occupants (7 × 80 kg average)	560	≈ 2.5 %
Design margin / contingency	1 900 – 2 100	≈ 9 %
Total Operational Mass (hard ceiling)	22 000	100 %

3. Six Independent Safety Layers – Detailed Description

The defining and differentiating feature of the Ultra-Safe Central Capsule is the architecture of six fully independent safety layers. Each layer is engineered to be capable of performing its critical function even if every preceding layer has failed completely. This eliminates single-point failures across the entire survival sequence: detection of the emergency condition, clean separation from the host vehicle, parachute deployment, controlled descent, impact attenuation, and post-landing survival and location. The layers are described below in the chronological order in which they are normally expected to act under a typical emergency sequence, but each layer remains independently capable and is not dependent on the successful operation of any earlier layer.

Layer 1 – Primary Electrical Command and Sensing

Dual-redundant processors continuously monitor a comprehensive suite of multi-axis accelerometers, rate gyroscopes, barometric and differential pressure sensors, and free-fall detection algorithms. The sensing suite is designed to recognise both rapid-onset events (for example mid-air collision, structural break-up, or sudden loss of host structural integrity) and more gradual but still critical conditions (for example uncontrolled descent, prolonged high sink rate, or progressive loss of host control authority). Manual activation is provided by guarded push-buttons that are reachable from both the pilot and co-pilot positions without requiring the occupants to leave their seats. Power for Layer 1 is drawn from the main emergency battery bus, which itself incorporates continuous health monitoring and automatic switch-over to backup sources. Under normal and most abnormal conditions Layer 1 is responsible for initiating host separation and parachute deployment. All critical commands are cross-checked between the two independent processors before any actuation is authorised, providing protection against single processor or software faults.

Layer 2 – Independent Pyrotechnic Path

A completely isolated pyrotechnic initiation architecture is powered by dedicated non-rechargeable thermal batteries. Full galvanic isolation from the main electrical system is maintained at all times so that common-mode electrical failures, electromagnetic pulse events, or total loss of the main battery bus cannot disable this path. Automatic arming occurs after a configurable timeout (typically three to five seconds) if Layer 1 has not confirmed successful separation or deployment. Layer 2 is fully capable of initiating both host-vehicle separation and parachute cluster deployment. The pyrotechnic devices are selected and qualified for the expected temperature, vibration, humidity and long-term storage-life environment of the intended host platforms. Status of the pyrotechnic path, including battery health and continuity, is continuously monitored while the capsule is installed in the host and is displayed on the host crew interface.

Layer 3 – Pure Mechanical Path (Zero Electronics)

High-force mechanical springs, steel cable runs and inertial masses provide a completely non-electrical and non-pyrotechnic activation path. A dedicated manual lever is positioned so that any of the seven occupants can reach and operate it without requiring special tools, sequences or prior training beyond basic familiarisation. In addition, pure inertial triggering occurs automatically under high negative acceleration even in the complete absence of human action. Because this layer contains no electronics and no energetic materials beyond the stored spring energy, it remains fully functional after total electrical power loss, electromagnetic pulse events, or complete failure of the pyrotechnic system. Layer 3 constitutes the ultimate non-energetic backup for separation and deployment initiation and is regarded as one of the highest-reliability elements of the entire architecture.

Layer 4 – High-Pressure Cold-Gas Deployment System

High-pressure bottles containing nitrogen or an equivalent inert gas are fitted with mechanical-thermal valves that require no electrical power to open once the thermal or mechanical trigger condition is met. If Layers 1 through 3 have not achieved confirmed parachute deployment, these valves open and the stored gas provides forced ejection and inflation assistance for the parachute cluster. The same gas system can supply supplementary force for host separation if required by the separation dynamics. Layer 4 is the final energetic deployment path and is designed to remain functional after prolonged storage periods and under extreme temperature conditions. Bottle pressure and valve health are continuously monitored while the capsule is installed in the host vehicle.

Layer 5 – Impact Energy Absorption and Structural Survival

Eight mechanical landing legs, evenly spaced around the circumference of the capsule, are each equipped with progressive hydraulic dampers and multi-rate elastomer pads. Viscoelastic elements in the seat suspensions provide an additional stage of energy absorption for the occupants. The continuous carbon-fiber floor and the full six-axis roll-cage (designed to a safety factor of at least 2.8 on primary members under ultimate combined loads) maintain a survivable internal volume under extreme multi-axis impact, roll-over and full inversion loads. Layer 5 is the last line of structural protection if the entire parachute system fails to deploy or is destroyed by external events. Its performance is verified by a combination of non-linear finite-element analysis and physical drop and impact testing at progressively higher energy levels, explicitly including off-axis and asymmetric impact cases that represent realistic accident conditions.

Layer 6 – Post-Landing Survival, Flotation and Location

After a successful or partially successful landing, Layer 6 maximises the probability of timely rescue. Automatically deployed flotation volumes are sized to support the full operational mass of 22 000 kg with an appropriate reserve of buoyancy and, together with the external geometry of the capsule, provide a self-righting capability that returns the capsule to an upright floating attitude after water impact or inversion. Dual-band emergency locators (406 MHz COSPAS-SARSAT beacon together with GNSS positioning and two-way satellite messaging) combined with high-intensity visual strobes and an acoustic beacon enable rapid location by search-and-rescue assets. A minimum 72-hour life-support package for seven occupants, comprising oxygen (stored gas and/or chemical generation), active CO₂ scrubbing, basic particulate and contaminant filtration, emergency water and calorie-dense rations, is carried. Dual-agent fire suppression remains available. The capsule is environmentally sealed against water ingress and external contaminants. Manual activation and override controls for all Layer 6 functions are available to the occupants at all times.

4. Structural Architecture – Continuous Carbon-Fiber Roll-Cage

The primary load-carrying structure is a continuous six-axis carbon-fiber roll-cage constructed from large-section primary tubes in the 15–18 cm class, joined by precision titanium node fittings at all primary intersections and attachment points. The entire floor plane is continuous and contains no hatch or other discontinuity. This design decision maximises torsional and bending stiffness and provides a rigid, unbroken foundation for the eight landing-leg attachment points distributed around the circumference.

Local reinforcement is provided around the two side-door apertures and the single roof hatch. Load paths originating at these openings are returned into the continuous cage network through the titanium nodes so that the global strength and stiffness of the cage are preserved. The target safety factor on primary members under ultimate combined loads (impact + roll + asymmetric landing) is greater than or equal to 2.8. Energy-absorbing interfaces between the cage and the seating system reduce the peak accelerations transmitted to the occupants. All external surfaces receive corrosion-resistant coatings suitable for long-term maritime and mixed-environment service.

The structural concept is intentionally simple and highly redundant. There is no reliance on single critical joints or on the integrity of any one door aperture. Even after significant local damage the remaining cage structure and the continuous floor are expected to maintain a survivable volume for the occupants. This philosophy of graceful degradation under damage is central to the overall safety case and is a locked design principle.

5. Parachute System and Host Separation

5.1 Parachute Cluster

The recovery parachute system consists of eight primary canopies supplemented by eight auxiliary or stabiliser canopies, giving a total of sixteen canopies. The cluster is sized for controlled descent of the full operational mass of 22 000 kg with appropriate margin. Each primary canopy is fitted with dual pilot/extractor chutes whose function is to force opening of the main canopy if it is fouled or delayed. Inflation-confirmation sensors monitor each canopy; if confirmation is not received within a short, predefined time window the next safety layer is automatically commanded. Both fully automatic and fully manual deployment modes are permanently available to the occupants. The system is designed so that the loss of one or more individual canopies still permits a survivable descent rate. Total canopy area, reefing schedule and deployment sequence are sized against the hard mass ceiling of 22 000 kg.

5.2 Host-Vehicle Separation System

Separation from the host is accomplished by parallel pyrotechnic and hydraulic release actuators. In addition, pure mechanical locks are provided that can be released by a manual lever even after total loss of electrical and hydraulic power. The separation plane geometry is arranged to minimise the risk of re-contact or entanglement with residual host structure. All electrical, data and fluid interfaces are designed for a clean break at the moment of separation. Pre-separation health monitoring and status indication are available on the host crew interface when the capsule is installed. The separation system is required to function under the full range of host attitudes and load conditions that can reasonably be expected at the moment an emergency is declared.

6. Impact Energy Management

Even in the worst-case scenario of complete parachute failure the capsule is required to give the occupants a high probability of survival. Energy management is achieved through a staged, progressive system that converts kinetic energy into controlled deformation and heat over a sufficiently long stroke. The sequence of energy absorption is as follows:

1. The eight mechanical legs deploy or extend to establish a stable landing attitude and to begin dissipating vertical kinetic energy.
2. Progressive hydraulic dampers absorb the majority of the remaining vertical energy over a controlled stroke.
3. Multi-rate elastomer pads provide the final stage of stroke and attenuate residual high-frequency shock.
4. Viscoelastic elements in the seat suspensions and the occupant restraint system further limit the peak loads experienced by the human body.
5. The continuous carbon-fiber floor and the roll-cage maintain a survivable internal volume under asymmetric, multi-axis and inversion loading.

The performance of the energy-absorption system is established by a combination of non-linear finite-element analysis and physical drop and impact testing at progressively higher energy levels. Off-axis and asymmetric impact cases are explicitly included in the test matrix because they represent realistic accident conditions that must be addressed by the design.

7. Life Support, Environmental Control and Fire Protection

After landing or water impact the occupants may have to wait an extended period for rescue. The capsule therefore carries a minimum 72-hour life-support capability for seven persons. Oxygen is provided by a combination of stored gas and/or chemical generation systems, together with active CO₂ scrubbing and basic particulate and contaminant filtration. Emergency water and calorie-dense rations are sized for the same duration. Passive insulation supplemented by limited active thermal control maintains internal temperatures within survivable limits across the expected range of external environmental conditions.

Fire protection employs a dual-agent architecture. A clean gaseous agent is used for the occupied interior volume so that the atmosphere remains breathable after discharge. A foam or powder capability is provided for external or residual fire threats. Coverage is designed to be omnidirectional. The capsule remains environmentally sealed after separation against water ingress and external contaminants. Fire detection is linked to the overall emergency logic so that an internal fire can, if necessary, contribute to the decision to separate from a burning host vehicle.

8. Flotation, Self-Righting and Emergency Location

For maritime and amphibious applications the capsule incorporates automatically deployed flotation volumes sized to support the full operational mass of 22 000 kg with an appropriate reserve of buoyancy. The external geometry, in combination with the flotation system, provides a self-righting capability that returns the capsule to an upright floating attitude after water impact or inversion. This capability is essential both for occupant comfort and for the proper operation of the location beacons, which are optimised for upright orientation.

Emergency location is provided by a dual-band system comprising a 406 MHz COSPAS-SARSAT beacon together with GNSS positioning and two-way satellite messaging. High-intensity visual strobes and an acoustic beacon further assist visual and auditory location by search-and-rescue teams. All location functions can be activated automatically upon separation or impact, or manually by the occupants at any time.

9. Door and Access Configuration

Access configuration was finalised to balance structural strength against practical boarding and emergency egress requirements. The locked configuration is as follows:

- **Two side doors** (port and starboard) serve as the primary emergency egress routes after separation and landing. They are sized and positioned for rapid exit by all seven occupants without requiring complex sequences.
- **One large reinforced roof hatch** is the primary boarding path while the capsule is installed in the host vehicle and the primary external rescue access after landing.
- **No floor hatch** is provided. The entire bottom plane remains continuous. This decision maximises the strength and stiffness available to the eight landing-leg attachment nodes and preserves the continuity of the roll-cage under the most severe impact and roll-over loads.

All apertures are locally reinforced. Load paths originating at the openings are returned into the primary cage structure through titanium nodes so that the global integrity of the roll-cage is not compromised.

10. Host Platform Interface Requirements

The capsule is conceived as a modular unit that can be integrated into multiple host platform types. The minimum interface requirements that any host must satisfy are the following:

- Structural hard-points capable of carrying the full 22 000 kg capsule mass under the complete range of host crash, flight and operational loads, with appropriate safety factors consistent with the host certification basis.
- A defined separation plane incorporating controlled mechanical, electrical, data and fluid disconnects that achieve a clean break at the moment of separation under all expected attitudes and load conditions.
- Provision for continuous pre-separation health monitoring and status display on the host crew interface, covering at minimum the status of Layers 1, 2 and 4 and the overall readiness of the capsule.
- Sufficient clearance volume around the capsule to permit clean separation without interference from residual host structure, taking into account the 5.80 m diameter by 3.80 m height envelope plus the required separation kinematics.
- An optional umbilical for continuous power and data while the capsule is installed; the umbilical shall be automatically severed or disconnected at the moment of separation.
- Host structure and systems shall not impose loads or environmental conditions on the capsule that exceed the design envelope defined in this baseline.

11. Scalability and Family Concept

The Ultra-Safe Central Capsule is designed as a scalable architecture. The large configuration defined in this document (5.80 m diameter × 3.80 m height, 22 000 kg controlling mass) forms the reference baseline. Smaller and larger members of the same family may be developed while preserving the same core principles: six independent safety layers, continuous carbon-fiber roll-cage with unbroken floor plane, multi-path separation and deployment, circular/radial seating philosophy, and post-landing survival systems. Scaling is expected to be performed primarily by changes in diameter, height, structural member sizing, parachute cluster capacity and landing-leg energy-absorption capacity, while retaining the same logical architecture and safety-layer independence rules. This scalability supports a coherent product family and a unified intellectual-property portfolio suitable for long-term licensing.

12. Future Variant Option – Concealed Rotors with Limited Hover Capability

The baseline configuration defined in this document is a pure survival capsule without propulsion or rotor systems. However, the architecture is intentionally structured so that future variants equipped with concealed rotors providing limited hover capability may be designed and engineered if requested by a licensee. Such variants would incorporate ducted or otherwise concealed rotor systems integrated into the lower structure or floor region, sized only for limited hover, soft landing assistance or short-duration station-keeping, and would remain subject to the same six independent safety layers, continuous roll-cage philosophy and mass-control discipline. Development of any powered hover variant shall be treated as a formal derivative configuration requiring its own mass budget, structural assessment, safety assessment and verification programme, and shall not silently alter the pure survival baseline defined herein. The option is recorded in this baseline solely to preserve design freedom for future licensed development and does not form part of the controlling pure-survival specification.

13. Verification and Validation Approach

A structured verification and validation programme is required before the design can be considered ready for certification or production release. Each requirement in the specification will ultimately be linked to one or more methods of verification (analysis, inspection, demonstration or test). The principal elements of the programme are outlined below.

- **Structural analysis and test** – Linear and non-linear finite-element analysis covering static, dynamic, impact and fatigue cases, correlated with physical static and dynamic testing of critical assemblies and of the complete cage.
- **Energy-absorption testing** – Full-scale or suitably scaled drop tests and impact tests of the eight-leg damper system at progressively higher energy levels, including off-axis and asymmetric cases.
- **Parachute deployment trials** – Instrumented deployment tests under representative dynamic conditions, including deliberate failure-injection cases to verify the operation of successive safety layers.
- **Separation testing** – Controlled separation tests from representative host structures to confirm clean release, interface break and absence of re-contact.
- **Flotation and self-righting demonstrations** – Water-basin or open-water tests confirming flotation reserve and self-righting behaviour after inversion.
- **Life-support endurance** – Closed-capsule tests at full occupant load verifying atmosphere control, thermal environment and consumable endurance for the specified 72-hour period.
- **Environmental qualification** – Temperature, vibration, humidity, salt-fog and electromagnetic compatibility testing appropriate to the intended operational environments.
- **Human-factors evaluation** – Assessment of seating, harnesses, control reach, emergency procedures and egress under representative conditions.

14. Preliminary Patent-Family Assessment

The six-layer architecture, the continuous-floor high-strength carbon-fiber cage, the multi-path separation and deployment logic, the integrated progressive energy-absorption system with eight landing legs, the flotation and self-righting capability, and the long-duration survival and location systems together support a broad and defensible patent portfolio.

Estimated number of extractable patent families: **18 – 28**

Of which judged high to very-high commercial and defensive strength: **12 – 18**

Strongest conceptual clusters identified to date:

- Six-layer independent safety architecture specifically adapted to multi-domain survival modules.
- Continuous-floor carbon-fiber roll-cage combined with multi-leg progressive energy absorption.
- Multi-path (electrical / pyrotechnic / pure mechanical / cold-gas) deployment and separation logic.
- Integrated flotation, self-righting and dual-band location system enabling rapid multi-domain recovery.
- Modular host-interface and clean-break separation architecture suitable for diverse platform types.
- Scalable family architecture preserving core safety principles across different size classes.

15. Key Specification Summary (Locked Values)

Item	Specification
Geometry	5.80 m diameter × 3.80 m height (cylindrical, circular planform)
Occupants	7 persons, circular / radial seating
Operational Mass	22 000 kg (hard ceiling / controlling target)
Primary Structure	Continuous CFRP roll-cage + continuous floor, safety factor ≥ 2.8
Landing Legs	8 progressive hydraulic + elastomer legs, evenly spaced
Safety Layers	Six independent layers (Electrical, Pyrotechnic, Mechanical, Cold-Gas, Impact Absorption, Post-Landing)
Parachute System	8 primary + 8 auxiliary canopies, multi-path deployment, sized for 22 t

Item	Specification
Impact Absorption	8 legs + progressive dampers + elastomer pads + viscoelastic seats
Life Support	≥ 72 hours for 7 occupants (O ₂ , CO ₂ scrubbing, water, rations)
Fire Suppression	Dual-agent (clean gas interior + foam/powder exterior), omnidirectional
Flotation & Location	Automatic flotation + self-righting + dual-band 406 MHz / satellite beacons
Access Configuration	2 side doors + 1 reinforced roof hatch; fully continuous floor (no floor hatch)
CG Envelope	Within 120 mm of geometric centre (vertical and horizontal)
Scalability	Family architecture; larger and smaller variants preserve core safety principles
Future Hover Option	Concealed-rotor limited-hover variants may be developed as formal derivatives if requested by licensee
Patent Potential	18–28 families (12–18 of high to very-high strength)

16. Document Control and Recommended Next Steps

This document (Document 00, Version 1.6 Full Detailed Engineering Baseline) constitutes the authoritative technical baseline for the Ultra-Safe Central Capsule Large Configuration. All subsequent detailed design, analysis, simulation, testing, certification planning, patent-extraction and licensing activities shall reference this baseline. Any material change to requirements, geometry, controlling mass, safety-layer architecture or primary structural concept shall be captured in a formal revision of this document and shall trigger appropriate re-verification.

Recommended immediate next actions:

- Freeze the external geometry (5.80 m diameter × 3.80 m height) and the controlling mass budget (22 000 kg hard ceiling); release them for detailed CAD modelling.
- Initiate high-fidelity finite-element models of the continuous roll-cage, continuous floor and eight leg-attachment regions.
- Develop detailed logic diagrams, timing sequences and FMEA for Safety Layers 1 through 4.
- Size the eight-primary-plus-eight-auxiliary parachute cluster and the eight-leg energy-absorption stroke for the hard mass ceiling of 22 000 kg.
- Prepare host-interface control drawings for the priority platform applications.
- Commission preliminary patent novelty searches on the strongest family clusters identified in Section 14.
- Draft the top-level verification matrix that links each requirement to analysis, inspection or test methods.
- If a licensee requests a limited-hover derivative, open a formal derivative configuration under change control without altering this pure-survival baseline.